

MACHINE LEARNING

LINEAR REGRESSION

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MASTER IN BUSINESS ANALYTICS



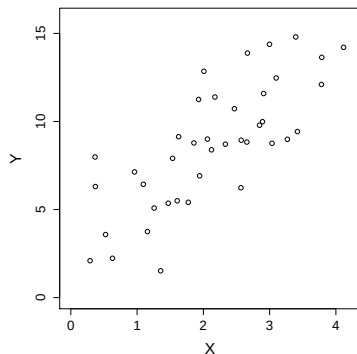
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Simple linear regression

- ▶ Recall: $Y = f(X) + \epsilon$
- ▶ Simple linear model:

$$f(X) = \beta_0 + \beta_1 X,$$

where β_0 and β_1 are model coefficients or parameters.



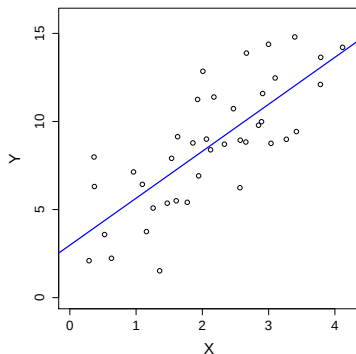
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- ▶ Fitted value: $\hat{y}_i = \hat{f}(x_i) = \hat{\beta}_0 + \hat{\beta}_1 x_i$



Simple linear regression

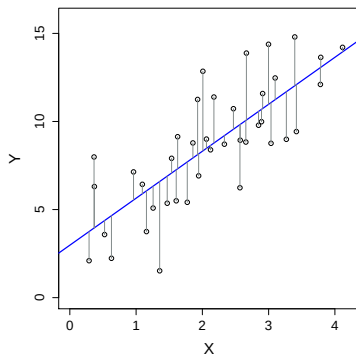
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- ▶ **Fitted value:** $\hat{y}_i = \hat{f}(x_i) = \hat{\beta}_0 + \hat{\beta}_1 x_i$
- ▶ **Residual:** $e_i = y_i - \hat{y}_i$
- ▶ **Residual sum of squares (RSS)**

$$\begin{aligned}\text{RSS}(\hat{\beta}_0, \hat{\beta}_1) &= e_1^2 + e_2^2 + \cdots + e_n^2 \\ &= \sum_{i=1}^n (y_i - \hat{f}(x_i))^2 \\ &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2\end{aligned}$$



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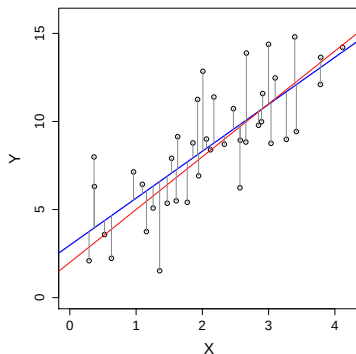
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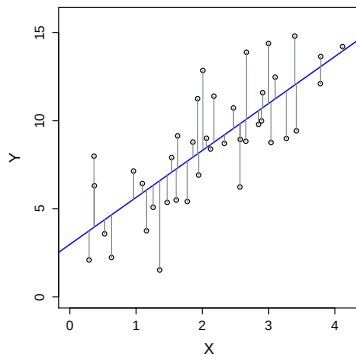
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Estimation of parameters by least squares

- To estimate (β_0, β_1) , we minimize the RSS:

$$\begin{aligned}(\hat{\beta}_0, \hat{\beta}_1) &= \operatorname{argmin}_{(\beta_0, \beta_1) \in \mathbb{R}^2} \operatorname{RSS}(\beta_0, \beta_1) \\ &= \operatorname{argmin}_{(\beta_0, \beta_1) \in \mathbb{R}^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2\end{aligned}$$



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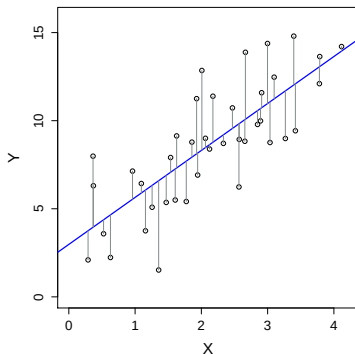
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- ▶ How do we solve this optimization problem?
- ▶ Numerically with computer algorithm (**gradient descent**, etc.)
- ▶ Analytically by differentiation (if possible):

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ and $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$
are **sample means**.



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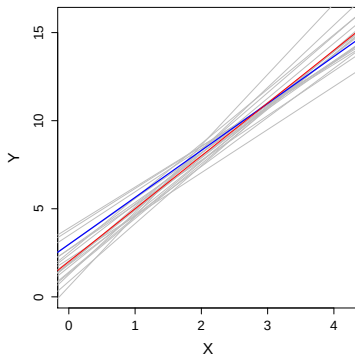
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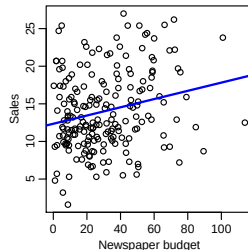
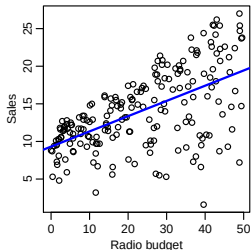
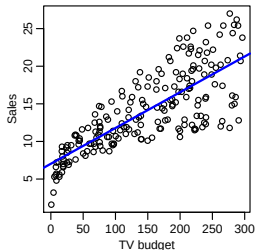
$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \\ \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x},\end{aligned}$$

where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ and $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ are **sample means**.

- ▶ Note: $(\hat{\beta}_0, \hat{\beta}_1)$ are **random**, since for different training data we obtain different estimates!



Advertising data



Coefficient	Estimate	Std. Error	<i>t</i> value	<i>p</i> -value
Intercept (β_0)	7.032594	0.457843	15.36	$< 2e-16$
TV (β_1)	0.047537	0.002691	17.67	$< 2e-16$

Coefficient	Estimate	Std. Error	<i>t</i> value	<i>p</i> -value
Intercept (β_0)	9.31164	0.56290	16.542	$< 2e-16$
radio (β_1)	0.20250	0.02041	9.921	$< 2e-16$

Coefficient	Estimate	Std. Error	<i>t</i> value	<i>p</i> -value
Intercept (β_0)	12.35141	0.62142	19.88	$< 2e-16$
newspaper (β_1)	0.05469	0.01658	3.30	0.00115

Multiple linear regression

- ▶ We consider p predictors $X = (X_1, \dots, X_p)$ for the response Y . The model is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \epsilon.$$

- ▶ For the **Advertising** data set:

$$\text{sales} = \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times \text{newspaper} + \epsilon.$$

- ▶ The training data $\{(x_i, y_i)\}_{i=1}^n$ with $x_i = (x_{i1}, \dots, x_{ip})^\top$ can be written as an $n \times p$ matrix

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}$$

- ▶ For notational convenience, we sometimes use the **dot product** between $x, y \in \mathbb{R}^p$

$$x^\top y = \sum_{i=1}^p x_i y_i = x_1 y_1 + \dots + x_p y_p.$$

Estimation and prediction in multiple linear regression

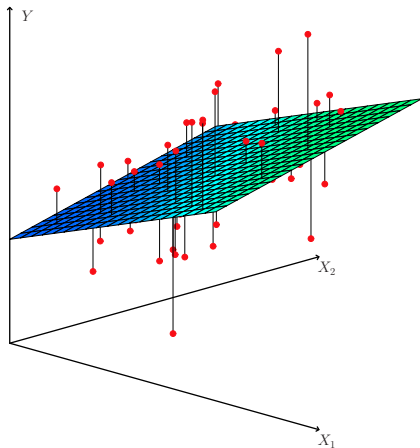
- As before, we obtain parameter estimates $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ as those parameters that minimize the **RSS**

$$\begin{aligned}\text{RSS}(\beta_0, \dots, \beta_p) &= \sum_{i=1}^n (y_i - \beta_0 - \mathbf{x}_i^\top \boldsymbol{\beta})^2 \\ &= \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_p x_{ip})^2\end{aligned}$$

- We make **predictions** at a new point $\mathbf{x}_0 = (x_{01}, \dots, x_{0p})$ by

$$\begin{aligned}\hat{y}_0 &= \hat{\beta}_0 + \mathbf{x}_0^\top \hat{\boldsymbol{\beta}} \\ &= \hat{\beta}_0 + \hat{\beta}_1 x_{01} + \dots + \hat{\beta}_p x_{0p},\end{aligned}$$

where $\hat{\boldsymbol{\beta}} = (\hat{\beta}_1, \dots, \hat{\beta}_p)$.



Advertising data

Coefficient	Estimate	Std. Error	<i>t</i> value	<i>p</i> -value
Intercept	2.939	0.3119	9.42	$< 2e-16$
TV	0.046	0.0014	32.81	$< 2e-16$
radio	0.189	0.0086	21.89	$< 2e-16$
newspaper	-0.001	0.0059	-0.18	0.8599

Correlations:

	TV	radio	newspaper
TV	1	0.0548	0.0567
radio		1	0.354
newspaper			1

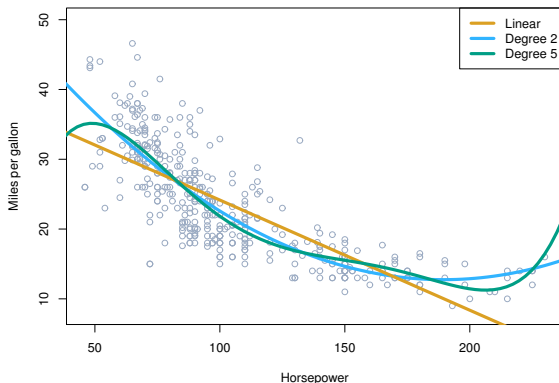
Example: the Auto data set

- ▶ The **Auto** data set contains measurements for $n = 397$ cars, such as **mpg** (miles per gallon), **horsepower**, **year**, etc.
- ▶ We can fit the model

$$\text{mpg} = \beta_0 + \beta_1 \times \text{horsepower} + \beta_2 \times \text{horsepower}^2 + \epsilon,$$

this is still a linear model (in $X_1 = \text{horsepower}$ and $X_2 = \text{horsepower}^2$).

- ▶ In this case, both β_1 and β_2 are significant.
- ▶ Even higher polynomials can be fitted.



Beyond linearity: linear basis functions

- ▶ Most often the true function $f(X)$ is not linear in the predictors $X_1, \dots, X_p$! But we can adjust the linear model.
- ▶ Instead of the $X = (X_1, \dots, X_p)$ we consider transformation of them. Let $h_m(X) : \mathbb{R}^p \rightarrow \mathbb{R}$ be the m th transformation, $m = 1, \dots, M$. The model is

$$f(X) = \sum_{m=1}^M \beta_m h_m(X).$$

- ▶ This model is **linear** in the new predictors, the **basis functions** $h_1(X), \dots, h_M(X)$; estimation and prediction can be applied as before!

Possible choices for h_m are:

- ▶ $h_m(X) = X_m$, $m = 1, \dots, p$, essentially recovers the original linear model.
- ▶ $h_m(X) = X_j^2$ or $h_m(X) = X_j X_k$ allows to achieve higher order polynomials.
- ▶ $h_m(X) = \log(X_j)$, $\sqrt{X_j}$ covers other non-linear transformation.
- ▶ etc.